ثانوية المدينة التعليمية
Education City High School

1.1

Change in Tandem

Core-to-Challenge Worksheet and AP Practice

AP Precalculus - Unit 1 | ECHS Mathematics | Teacher: Mohammad Abu-Ghuwaleh

Student Information

Name: ______________________ Class: __________ Date: __________

Directions

Show mathematical evidence. Retain domain restrictions, label graphs, include units, and write complete interpretations. Calculator use is permitted only when the item explicitly requires approximation, regression, or numerical solving.

Readiness

1. Write one precise sentence explaining this principle: A function assigns exactly one output to each input in its domain.

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2. Write one precise sentence explaining this principle: Domain restrictions may be mathematical, contextual, or both.

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3. Write one precise sentence explaining this principle: A repeated output is allowed; a repeated input with two different outputs is not.

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4. Write one precise sentence explaining this principle: Image means output produced by a stated input; preimage means input producing a stated output.

Core

5. A tank contains $V(t) = 120 - 8t$ liters for $0 \le t \le 15$. Find the domain and range and interpret $V(6)$.

6. Does $x^2 + y^2 = 9$ define y as a function of x on $[-3, 3]$?

7. For $f(x) = \sqrt{5 - 2x}$, determine the domain and find all preimages of 1.

8. State what the formula $f(a) =$ the output when the input is a tells you and name any required restriction.

9. State what the formula Dom$(f) = \{x : f(x)$ is defined$\}$ tells you and name any required restriction.

10. State what the formula Range$(f) = \{f(x) : x \in$ Dom$(f)\}$ tells you and name any required restriction.

Medium

11. Determine whether $\{(-2, 1), (0, 3), (2, 1), (0, 5)\}$ defines a function. Justify.

12. For $f(x) = \dfrac{1}{\sqrt{x - 2}}$, find the domain.

13. A ride costs $C(n) = 18 + 7n$ QAR for integer $n \in \{0, 1, \ldots, 6\}$. State domain and range.

14. For $f(x) = |x - 4|$, find all preimages of 6.

15. Explain why a vertical line graph $x = 3$ is not y as a function of x, but can define x as a function of y.

Hard

16. Find the range of $f(x) = x^2 - 6x + 11$ on $[1, 7]$.

17. Find the domain of $g(x) = \sqrt{\dfrac{x-1}{x+3}}$.

18. A table gives $f(-1) = 2, f(0) = 2, f(1) = 5, f(2) = 9$. Find the image of $\{-1, 0, 2\}$ and the preimage of 2.

19. Let $f(x) = \dfrac{x+1}{x^2-4}$. State its maximal real domain and explain why simplifying is impossible.

Difficult

20. Find all values of k for which the relation $y^2 - 4y = x + k$ defines exactly one y-value when $x = 5$.

21. Find the range of $f(x) = \dfrac{x-1}{x+2}$ without graphing.

22. A function has domain $[-2, 5]$, is increasing on $[-2, 1]$, decreasing on $[1, 5]$, and satisfies $f(-2) = 3, f(1) = 9, f(5) = -1$. What range is forced?

Challenge / HOT

23. Construct two different functions with domain $\{-1, 0, 1\}$ and range $\{2, 5\}$, one injective if possible and one not injective. Explain.

24. Prove that a relation and its inverse are both functions exactly when the original function is one-to-one.

AP-Style Multiple Choice

25. Which relation is a function?
- **(A)** $\{(1,2),(1,3)\}$
- **(B)** $x^2+y^2=1$
- **(C)** $y=\sqrt{x+2}$
- **(D)** $x=|y|$

26. The domain of $\sqrt{7-2x}$ is
- **(A)** $(-\infty,7/2]$
- **(B)** $[7/2,\infty)$
- **(C)** $(-\infty,7/2)$
- **(D)** $\mathbb{R}$

27. If $f(2)=7$, then 2 is
- **(A)** an image of 7
- **(B)** a preimage of 7
- **(C)** the range
- **(D)** the codomain

28. Which is allowed for a function?
- **(A)** one input with two outputs
- **(B)** two inputs with the same output
- **(C)** no domain
- **(D)** a vertical line containing multiple graph points

29. The range of x^2+1 on all real numbers is
- **(A)** $\mathbb{R}$
- **(B)** $[0,\infty)$
- **(C)** $[1,\infty)$
- **(D)** $(1,\infty)$

30. Which condition makes $x^2+y^2=16$ a function of x?
- **(A)** $x\geq 0$
- **(B)** $y\geq 0$
- **(C)** $x\neq 0$
- **(D)** $y\neq 0$

AP-Style Free Response

31. A drone altitude is modeled by $h(t)=-5t^2+30t+40$ for $0\leq t\leq 7$. (a) State domain and range. (b) Interpret $h(2)$. (c) Find all preimages of 60.

32. Let $f(x) = \sqrt{9 - x^2}$. (a) State domain and range. (b) Explain geometrically why it is a function. (c) Find the preimages of $\sqrt{5}$. (d) Extend the relation to the full circle and explain why the extension is not a function of x.

Reflection

Identify one item where a domain restriction, unit, assumption, or representation changed your reasoning. Explain in 2-3 sentences.